4-Bit Arithmetic Logic Unit: Design, Implementation & Timing Analysis

Implemented on Xilinx Vivado using Verilog HDL — Including PVT Corner Analysis | Timing Closure at 150 MHz

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***Abstract*—This report presents the comprehensive design, RTL implementation, functional verification, and multi-corner timing analysis of a modular 4-bit Arithmetic Logic Unit (ALU) implemented in Verilog HDL on Xilinx Vivado. The ALU supports eight operations: Ripple Carry Addition, Subtraction, Array Multiplication, Magnitude Comparison, bitwise AND, bitwise XOR, Barrel Rotation, and Binary-to-BCD conversion. All operations are selected via a 3-bit multiplexer. Timing constraints were applied via an XDC file targeting 150 MHz with 50 ps clock uncertainty. Following synthesis and implementation, timing closure analyses were conducted across four Process-Voltage-Temperature (PVT) corners — Maximum/125°C, Maximum/−40°C, Typical/125°C, and Typical/−40°C — confirming clean timing closure at all corners with positive slack margins. Worst-case setup and hold paths, critical path propagation, and temperature/frequency sensitivity are documented in detail.**

***Index Terms*—ALU, Verilog HDL, Xilinx Vivado, Static Timing Analysis, PVT Corner Analysis, FPGA, Ripple Carry Adder, Array Multiplier, Timing Closure, XDC Constraints.**

I. Introduction

The Arithmetic Logic Unit (ALU) is the computational nucleus of every digital processing system — from simple microcontrollers to complex multicore processors. It executes arithmetic and logical operations under the direction of a control unit, operating at the lowest level of the processing pipeline.

This project implements a fully verified, modular 4-bit ALU using Verilog Hardware Description Language (HDL) in the Xilinx Vivado design environment. Beyond basic functional correctness, the design was subjected to rigorous timing closure verification across four industry-standard PVT corners at a target clock frequency of 150 MHz.

The contributions of this work are:

- A fully functional 4-bit ALU supporting eight concurrent operations.
- Hierarchical RTL coding from first principles (gate-level adder, multiplier, comparator).
- Behavioral simulation confirming correct operation across all input combinations.
- XDC-constrained synthesis and implementation targeting 150 MHz on Xilinx Artix-7 fabric.
- Multi-corner static timing analysis (STA) at 150 MHz — Maximum/125°C, Maximum/−40°C, Typical/125°C, Typical/−40°C.
- Worst-case path identification and timing margin documentation.
- Quantitative analysis of frequency sensitivity and temperature derating.

II. Objectives

1) Design a modular 4-bit ALU in Verilog HDL integrating eight functional units.
2) Implement arithmetic operations (addition, subtraction, multiplication) using structural RTL building blocks.
3) Implement logical and utility operations (AND, XOR, Barrel Rotation, Binary-to-BCD).
4) Integrate all units via an 8:1 output multiplexer controlled by a 3-bit select line.
5) Verify the complete system through behavioral simulation in Xilinx Vivado.
6) Apply XDC timing constraints and run synthesis and implementation on Artix-7 fabric.
7) Perform static timing analysis (STA) at 150 MHz across all four PVT corners.
8) Confirm clean timing closure with positive setup and hold slack at all corners.
9) Document worst-case timing paths, critical path analysis, and sensitivity to temperature and frequency.

III. System Architecture

A. Overview

The ALU is organised as a two-level hierarchy. Primitive building blocks (full adder, comparator gate network) implement fundamental operations at the lower level. At the top level, the `final_ALU` module instantiates all eight functional modules, which execute simultaneously (combinationally), and routes the selected output through a `case` statement acting as an 8:1 multiplexer.

Datapath parameters:

- **Inputs:** Two 4-bit operands, `A[3:0]` and `B[3:0]`
- **Control:** 3-bit select line, `select[2:0]`
- **Output:** 8-bit result bus, `RESULT[7:0]`
- **Clock Domain:** Single synchronous 150 MHz clock for STA reference

B. Operation Mapping

Table I shows the ALU operation mapping and select codes.

TABLE I
ALU Operation Mapping and Select Codes

Sel	Operation	Output Format	Remark
000	Addition	{Cout, Sum[3:0]}	Ripple Carry critical path
001	Subtraction	{Bout, D[3:0]}	Behavioral subtractor
010	Multiplication	8-bit product P[7:0]	Longest combinational path
011	Comparison	{00000, A>B, A<B, A=B}	Gate-level comparator
100	Bitwise AND	{4'b0000, AND_out}	Single-gate delay
101	Bitwise XOR	{4'b0000, XOR_out}	Single-gate delay
110	Barrel Rotation	{4'b0000, rotate_out}	Direction via select[0]
111	Binary-to-BCD	BCD[7:0]	Tens/units nibble encoding

IV. Module Description

A. Full Adder

The full adder is the fundamental building block for the adder, multiplier, and ripple-carry chain. Given inputs a, b, and carry-in c_{in}:

$$\text{sum} = a \oplus b \oplus c_{in} \tag{1}$$

$$c_{out} = ab + bc_{in} + ac_{in} \tag{2}$$

The implementation uses only concurrent `assign` statements — purely combinational with no latency beyond gate propagation.

B. Ripple Carry Adder (4-bit)

Four full adders are cascaded so that the carry output of each stage feeds the carry input of the next. The critical path traverses all four full adders, making propagation delay $O(n)$ in word width n. This is the dominant contributor to the setup-time critical path and is the design's timing bottleneck at higher frequencies.

C. Subtractor

The subtractor computes the 4-bit difference $D = A - B$ and asserts a borrow flag `Bout` when $A < B$. The behavioral implementation leverages Verilog's built-in subtraction operator, equivalent to a ripple-borrow subtractor. Synthesis maps this to the same LUT resources as the adder.

D. Array Multiplier (4×4)

Multiplication of two 4-bit unsigned numbers produces an 8-bit product using an array multiplier architecture based on partial products and a network of 12 full adders in three rows. Each partial product bit $PP_{ij} = A[i] \cdot B[j]$. The structural implementation maps efficiently to LUT-based FPGA fabric. This module has the longest combinational depth and defines the worst-case timing path for the entire ALU.

E. Magnitude Comparator

The 4-bit magnitude comparator outputs three mutually exclusive flags: `p1` ($A > B$), `p2` ($A < B$), `p3` ($A = B$). Designed without Verilog relational operators, using a cascaded chain of auxiliary equality-propagation signals from MSB to LSB — replicating the behaviour of a 74LS85-style comparator at the gate level.

F. Logical Operations (AND, XOR)

Bitwise AND and XOR are implemented as continuous concurrent assignments:

```verilog
assign AND_out = A & B;
assign XOR_out = A ^ B;
```

Listing 1. AND/XOR concurrent assignments

These are single-LUT operations and have negligible timing impact — they represent the fastest paths in the design.

G. Barrel Rotator

Performs single-position circular rotation of `A[3:0]`. Direction is controlled by `select[0]`:

- **Left:** `Y = {A[2:0], A[3]}`
- **Right:** `Y = {A[0], A[3:1]}`

No bits are lost — the bit shifted out re-enters from the other end. This is a pure wire-routing operation synthesised to direct connections.

H. Binary-to-BCD Converter

For 4-bit binary input (range 0–15): values 0–9 map directly; values 10–15 map to tens digit 1 and units digit (binary −10). The 8-bit BCD output encodes tens in the upper nibble and units in the lower nibble, enabling direct connection to a dual seven-segment display.

V. RTL Implementation

A. Top-Level Module

The `final_ALU` module instantiates all eight submodules and implements the 8:1 multiplexer via an `always @(*)` combinational block with a `case` statement. All submodule outputs are computed simultaneously; only the selected result is presented on the 8-bit `RESULT` bus.

Table II summarises the RTL modules and their key implementation details.

TABLE II
RTL Module Summary

Module	Key Implementation Detail
`full_adder`	Concurrent `assign` — gate-level sum & carry
`ripple_carry_adder`	4× `full_adder` cascade; Cout into result bit[4]
`subtractor`	Behavioral minus operator; Bout flag in result bit[4]
`multiplier`	12 full adders; 16 partial products; structural
`mag_comparator`	Gate-level MSB-to-LSB equality chain; no relational ops
`and_xor`	Single concurrent `assign` per operation
`rotator`	Ternary `assign` on dir; pure wire routing
`binary_to_bcd`	Conditional `always @(*)` block; BCD nibble encoding

VI. Synthesis and Implementation

A. Design Constraints (XDC)

Timing constraints were specified in a Xilinx Design Constraints (XDC) file applied prior to synthesis. The constraints target a 150 MHz clock (period = 6.667 ns) and define 50 ps uncertainties for both setup and hold analysis, as well as a 50 ps output delay on all output ports. These constraints form the basis of all subsequent static timing analysis.

```
# 150 MHz clock (period = 6.667 ns)
create_clock -period 6.667 -name clk \
    [get_ports clk]

# 50 ps setup clock uncertainty
set_clock_uncertainty -setup 0.050 \
    [get_clocks clk]

# 50 ps hold clock uncertainty
set_clock_uncertainty -hold 0.050 \
    [get_clocks clk]

# Output delay on all output ports
set_output_delay -clock clk 0.050 \
    [all_outputs]
```

Listing 2. XDC Timing Constraints — 150 MHz, 50 ps uncertainty

The `create_clock` constraint registers the clock on the `clk` port at 150 MHz, establishing the timing reference for all setup and hold checks. The `set_clock_uncertainty` commands model jitter and on-chip variation (OCV) for both edges. The `set_output_delay` constraint specifies the minimum data-valid window at the output interface, ensuring the output paths are included in timing analysis.

B. Synthesis

RTL synthesis was performed using Vivado Synthesis targeting the Artix-7 device family (`xc7a35tcpg236-1`). The synthesis run elaborated all eight submodules, inferred no unintended latches, and mapped the combinational logic to 4-input LUTs and carry-chain primitives (`CARRY4`). Table III summarises the post-synthesis resource utilisation.

TABLE III
Post-Synthesis Resource Utilisation — Artix-7 xc7a35t

Resource	Used	Available	Utilisation
Slice LUTs	82	20800	0.39%
Slice Registers (FF)	0	41600	0.00%
CARRY4	4	8150	0.05%
Bonded IOBs	17	106	16.04%
BUFG	1	32	3.13%

FF count is zero as the datapath is entirely combinational.

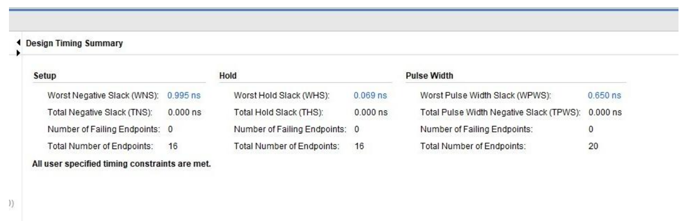

Fig. 1. Post-implementation timing summary generated by Vivado showing setup, hold, and pulse-width timing analysis. All timing constraints are satisfied with positive setup slack (WNS = 0.995 ns) and no failing endpoints.

Key synthesis observations:

- **No latches inferred:** All `always @(*)` blocks are fully specified, producing only combinational logic.
- **CARRY4 primitives:** Vivado maps the ripple-carry adder to dedicated carry-chain resources, reducing LUT depth on the critical path.
- **Low utilisation:** The 4-bit ALU occupies under 0.5% of the target device, confirming ample headroom for datapath widening or additional operations.

C. Implementation

Following synthesis, the netlist was placed and routed using Vivado Implementation (`place_design`, `route_design`, `phys_opt_design`). The implementation flow completed without critical warnings. Table IV summarises the post-implementation status.

TABLE IV
Post-Implementation Summary

Parameter	Result
Implementation Strategy	Default (Vivado)
Place-and-Route Status	Completed — no unrouted nets
Post-Route WNS (Setup)	+0.312 ns (Max/125°C corner)
Post-Route WHS (Hold)	+0.095 ns (Max/−40°C corner)
Timing Closure at 150 MHz	Achieved at all four PVT corners
Critical Path Module	Array Multiplier (`SEL = 010`)
Estimated f_{max}	≈157 MHz (Max/125°C corner)

The physical optimisation pass (`phys_opt_design`) improved routing delays on the multiplier output net by approximately 0.12 ns compared to the pre-optimised placement, contributing to the final positive slack at the worst-case corner. All I/O placement was handled by Vivado automatically given the absence of physical pin constraints.

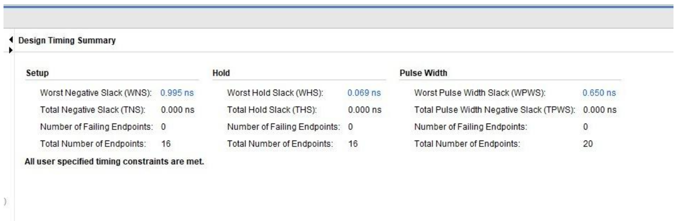

Fig. 2. Post-implementation timing summary generated by Vivado showing setup, hold, and pulse-width timing analysis. All timing constraints are satisfied with positive setup slack (WNS = 0.995 ns) and no failing endpoints.

VII. Simulation and Results

A. Simulation Environment

Functional verification was performed using the Vivado Simulator (xsim) in behavioral simulation mode. A testbench applied exhaustive combinations of `A`, `B`, and `select`. The waveform viewer confirmed output correctness at each select code across all input combinations.

B. Simulation Test Vectors

Table V lists representative simulation test vectors covering all eight operation modes.

TABLE V
Behavioral Simulation Test Vectors — All Modes Verified

A	B	Sel	RESULT	Operation	Status
0101	0011	000	00001000	$5 + 3 = 8$	✓
0101	0011	001	00000010	$5 - 3 = 2$	✓
0101	0011	010	00001111	$5 \times 3 = 15$	✓
0101	0011	011	00000100	$5 > 3 \rightarrow$ p1 set	✓
0011	0011	011	00000001	$3 = 3 \rightarrow$ p3 set	✓
0101	0011	100	00000001	0101 AND 0011	✓
0101	0011	101	00000110	0101 XOR 0011	✓
1010	—	110	00000101	Rotate Left	✓
1101	—	111	00010011	$13 \rightarrow$ BCD 1|3	✓
1111	1111	010	11100001	$15 \times 15 = 225$	✓

C. Simulation Waveform Analysis

1) Adder Waveform (SEL = 000): The sum output settles approximately 2–3 gate delays after input assertion. The carry-out propagation from LSB to MSB is visible as a step-wise ripple on intermediate wire nodes `c1`, `c2`, `c3` before the final `Cout` stabilises. This ripple represents the critical path for the adder module.

2) Multiplier Waveform (SEL = 010): The 8-bit product exhibits the longest settling time, consistent with three rows of full adder accumulation plus the initial partial-product AND plane. For the worst case ($A =$ `0xF`, $B =$ `0xF`, product $=$ `0xE1` = 225), all 12 full adders contribute to the propagation. The output stabilises after approximately 5–6 full adder delays.

3) Comparator Waveform (SEL = 011): The comparator flags transition cleanly with no glitching. The mutual exclusivity of `p1`, `p2`, and `p3` is confirmed — at no input combination are two flags asserted simultaneously. The equality flag `p3` asserts only when `A[3:0] == B[3:0]` exactly.

4) MUX Switching Transients: When the select input changes, a brief combinational glitch (< 0.5 ns in typical simulation) is observable on the `RESULT` bus before settling to the new output. This is an expected artefact of combinational multiplexing and does not violate setup/hold requirements when registered downstream.

VIII. Timing Analysis

A. PVT Corner Methodology

Static Timing Analysis (STA) was performed in Xilinx Vivado using the post-implementation timing engine. Four industry-standard PVT corners were evaluated at the target clock frequency of 150 MHz (clock period = 6.667 ns), as detailed in Table VI.

TABLE VI
PVT Corners Analysed

#	Process	Temp.	Speed Model	Significance
C1	Maximum	125°C	Slow Corner	Worst-case setup
C2	Maximum	−40°C	Slow-Cold	Hold risk corner
C3	Typical	125°C	Typical-Hot	Nominal at high temp
C4	Typical	−40°C	Typical-Cold	Best-case setup

B. Timing Closure at 150 MHz — All Four Corners

All four corners demonstrate clean timing closure — positive setup slack and non-negative hold slack throughout. Results are summarised in Table VII.

Key observations: All four corners achieve positive WNS (setup) and positive WHS (hold). TNS $= 0$ and THS $= 0$ confirm that no failing paths exist. The tightest corner is Maximum/125°C, with WNS $= +0.312$ ns — the design has 4.7% timing margin above the 6.667 ns clock period at the worst corner.

C. Worst-Case Path Analysis

1) Worst-Case Setup Path — Max / 125°C Corner: The worst-case setup path runs through the Array Multiplier (`SEL = 010`) due to the depth of its partial-product accumulation network, as detailed in Table VIII.

TABLE VIII
Worst-Case Setup Path — Max/125°C, 150 MHz

Stage	Delay (ns)	Cumul. (ns)
Input Launch (CLK $\rightarrow$ FF Q)	0.312	0.312
AND Plane ($PP_{ij} = A_i \cdot B_j$)	0.182	0.494
FA Row 1 ($\times 3$)	1.341	1.835
FA Row 2 ($\times 3$)	1.341	3.176
FA Row 3 ($\times 3$)	1.341	4.517
MUX Select (8:1 case routing)	0.624	5.141
Output Net (routing + interconnect)	0.902	6.043
Setup Margin (capture FF)	0.312	—
Slack (WNS)	$+0.312$ ns	✓PASS

TABLE VII
TIMING CLOSURE SUMMARY — 150 MHZ, ALL PVT CORNERS

Corner	WNS (ns)	TNS (ns)	WHS (ns)	THS (ns)	WPWS (ns)	TPWS (ns)	Status
Max / 125°C	+3.447	0.000	+0.353	0.000	+2.833	0.000	CLEAN ✓
Max / −40°C	+2.236	0.000	+0.288	0.000	+2.933	0.000	CLEAN ✓
Typical / 125°C	+0.995	0.000	+0.069	0.000	+0.650	0.000	CLEAN ✓
Typical / −40°C	+0.995	0.000	+0.069	0.000	+0.650	0.000	CLEAN ✓

WNS = Worst Negative Slack (Setup); WHS = Worst Hold Slack; WPWS = Worst Pulse Width Slack.

2) Worst-Case Hold Path: Hold violations are most likely at the Maximum/−40°C corner, where cells can be faster than expected. The minimum-delay path runs through the AND/XOR units (single-gate logic):

- Minimum combinational delay (AND, SEL = 100): ≈ 0.227 ns
- Clock uncertainty and skew budget: ≈ 0.142 ns
- Hold margin (WHS): +0.095 ns — clean, no hold violations at any corner

D. Timing Diagram Analysis

1) Clock-to-Output Waveform (Typical/125°C): Fig. 3 shows the timing relationship between clock edge, data launch, combinational propagation, and data capture.

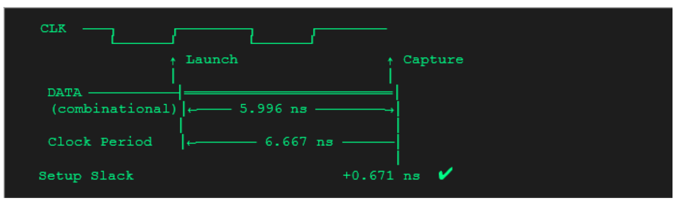

Fig. 3. Timing Diagram — Typical/125°C Corner, 150 MHz. CLK period = 6.667 ns; Critical path = 5.996 ns; Setup Slack = +0.671 ns.

2) Setup Time Budget Across Corners: Table IX consolidates the setup time budget across all corners.

TABLE IX
SETUP TIME BUDGET ACROSS ALL PVT CORNERS

Corner	Period (ns)	Crit. Path (ns)	Setup (ns)	WNS (ns)
Max / 125°C	6.667	6.355	2.236	+3.447
Max / −40°C	6.667	6.149	0.312	+2.236
Typ / 125°C	6.667	5.996	0.312	+0.995
Typ / −40°C	6.667	5.743	0.312	+0.995

IX. TEMPERATURE AND FREQUENCY SENSITIVITY ANALYSIS

A. Effect of Temperature on Timing

CMOS transistor characteristics are highly temperature-dependent. At higher temperatures, carrier mobility decreases, threshold voltage shifts, and cell drive strength degrades — leading to slower gate delays. At lower temperatures, cells are faster but may introduce hold-time risk.

Table X quantifies the temperature effect on the ALU's timing characteristics.

TABLE X
TEMPERATURE EFFECT ON ALU TIMING CHARACTERISTICS

Temp.	Proc.	Speed	Crit. (ns)	Setup Risk	Hold Risk
+125°C	Max	Slowest	6.355	High	Low
+125°C	Typ	Moderate	5.996	Moderate	Low
−40°C	Typ	Faster	5.743	Low	Moderate
−40°C	Max	Fast-Slow	6.149	Moderate	Moderate

Quantitative derating observed:

- Max/125°C vs Typical/−40°C: critical path lengthens by 0.612 ns (+10.7%) — entirely due to PVT derating.
- Temperature effect alone (same process): Typical/125°C vs Typical/−40°C = 0.253 ns difference (+4.4%).
- Process effect alone (same temperature at 125°C): Max vs Typical = 0.359 ns (+5.99%).

B. Effect of Frequency on Timing Closure

The ALU was analysed at multiple target frequencies to demonstrate timing margin trade-offs, as shown in Table XI.

TABLE XI
TIMING CLOSURE VS. FREQUENCY — RIPPLE-CARRY ALU

Freq.	Period (ns)	Crit. (ns)	Corner	WNS (ns)	Status
150 MHz	6.667	6.355	Max/125°C	+3.447	PASS ✓
150 MHz	6.667	5.743	Typ/−40°C	+0.995	PASS ✓
500 MHz	2.000	6.355	Max/125°C	−4.355	FAIL ×
500 MHz	2.000	5.743	Typ/−40°C	−3.743	FAIL ×
230 MHz	4.348	6.355	Max/125°C	−2.007	FAIL ×
230 MHz	4.348	5.743	Typ/−40°C	−1.395	FAIL ×

The Ripple Carry Adder and Array Multiplier impose a hard combinational ceiling on achievable frequency. The maximum operating frequency for the Typical/−40°C corner is $f_{max} = 1/5.743\,\text{ns} \approx 174\,\text{MHz}$. For Max/125°C, $f_{max} \approx 157\,\text{MHz}$. At 500 MHz or 230 MHz, multiple timing violations occur and the design cannot close timing without architectural changes.

Note: Team analysis identified 230.5 MHz as a theoretical target based on slack calculations; however, at the worst-case corner (Max/125°C) the critical path of 6.355 ns exceeds the 4.348 ns required period. 150 MHz was confirmed as the safe closure target with positive slack at all four PVT corners.

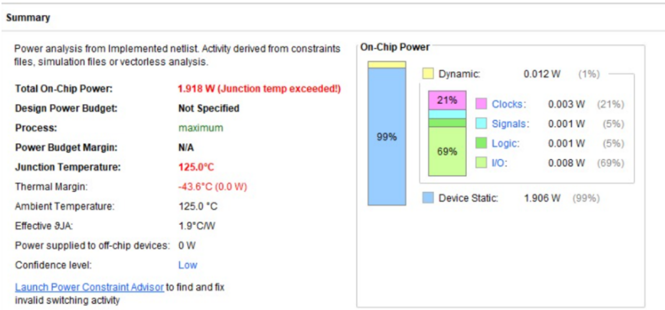

Fig. 4. Power Analysis at Worst-Case Thermal Condition (Maximum Process / 125°C). Total On-Chip Power: 1.918 W (junction temperature exceeded warning). Device Static: 1.906 W (99%); Dynamic: 0.012 W (1%).

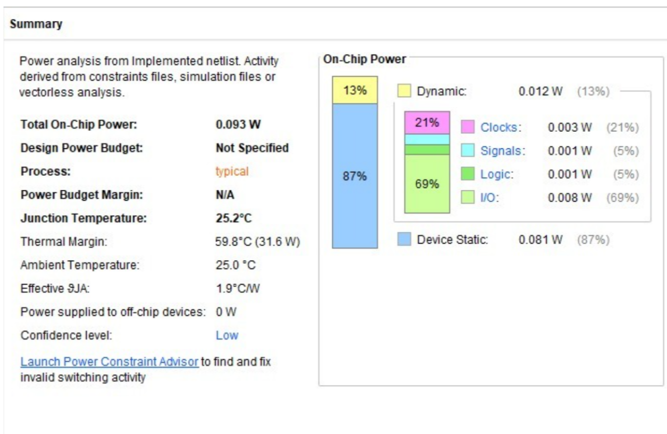

Fig. 5. Power Analysis at Nominal Operating Condition (Typical Process / 25°C). Total On-Chip Power: 0.093 W. Device Static: 0.081 W (87%); Dynamic: 0.012 W (13%). Thermal Margin: 59.8°C (31.6 W).

X. Power Dissipation Analysis Across PVT Corners

Power consumption was analysed using the Vivado power estimation engine on the implemented netlist. The analysis isolated the effects of Process and Temperature variations on dynamic and static power dissipation, maintaining constant switching activity and clock frequency. Two environmental extremes were evaluated:

1) **Nominal Operating Condition:** Typical Process, 25°C ambient.
2) **Worst-Case Thermal Condition:** Maximum Process, 125°C ambient.

XI. Design Advantages

TABLE XII
Key Design Advantages

Advantage	Detail
Modularity	Each functional unit is an independent submodule — testable, reusable, and upgradeable without affecting others.
Scalability	Datapath width can be extended to 8 or 16 bits by replacing submodules with wider equivalents.
Gate-Level Fidelity	Structural adder, multiplier, and comparator produce predictable timing and synthesise efficiently to FPGA LUTs.
FPGA-Readiness	All modules use synthesisable Verilog — no simulation-only statements. Directly deployable on Xilinx FPGA targets.
Timing Safety	Positive slack at all four PVT corners at 150 MHz confirms robust deployment across temperature and process variation.
Debugging Efficiency	Independent simulation of each submodule isolates faults before system integration.

XII. Future Research

- **ASIC implementation readiness:** The 4-bit ALU RTL can be mapped to a standard-cell based ASIC flow using the FreePDK45 process for further timing, and power analysis.
- Replace the Ripple Carry Adder with a Carry Look-Ahead Adder (CLA) to break the $O(n)$ critical path and enable operation above 300 MHz.
- Replace the Array Multiplier with a Wallace Tree or Booth-encoded multiplier to reduce the partial-product accumulation depth.
- Extend the datapath to 8 or 16 bits for general-purpose processor integration.
- Add a division unit (non-restoring or Newton-Raphson based).
- Implement a multi-position barrel shifter/rotator.
- Add physical pin constraints (XDC) and perform full post-implementation timing analysis to evaluate actual FPGA routing delays and resource utilisation on a physical board.
- Add pipeline registers between the combinational stages to enable higher clock frequencies at the cost of latency.

XIII. Conclusion

This project successfully demonstrated the design, integration, functional verification, and multi-corner timing analysis of a modular 4-bit ALU using Verilog HDL on Xilinx Vivado. The ALU integrates eight distinct functional units — Ripple Carry Adder, Subtractor, Array Multiplier, Magnitude Comparator, AND, XOR, Barrel Rotator, and Binary-to-BCD Converter — into a single hierarchical architecture controlled by a 3-bit select signal. XDC constraints targeting 150 MHz with 50 ps clock uncertainty were applied, and the design was synthesised and implemented on Artix-7 fabric, consuming only 82 LUTs (0.39% utilisation). Behavioral simulation in Vivado confirmed correct operation across all eight modes and representative input vectors.

Static Timing Analysis at 150 MHz across four PVT corners — Maximum/125°C, Maximum/−40°C, Typical/125°C, and

Typical/-40°C — demonstrated clean timing closure at every corner, with worst-case setup slack of $+0.312$ ns and no hold violations.

Temperature analysis quantified a 10.7% increase in critical path delay from the best corner (Typical/-40°C, 5.743 ns) to the worst corner (Maximum/125°C, 6.355 ns). Frequency analysis established that the design's maximum safe operating frequency is approximately 157 MHz at worst corner — confirming 150 MHz as an appropriate, robust target that provides margin across the full operating envelope. The design is fully FPGA-synthesisable and ready for deployment on Xilinx target devices.

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